AAE 251: Introduction to Aerospace Design

Course Syllabus

Spring 2026

Instructors:

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Graduate Teaching Assistants (GTA):

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Wyatt Donley, Veer Bhanchawat, Matiana De Alba, and Allen Kendall.

1 Interaction Opportunities

For course-related questions, you are encouraged to meet directly with the instructors:

Piazza Discussion Forum

You should already be enrolled in the **Piazza AAE 251 forum**. Upon enrollment, you will have received instructions on how to post questions. Please use Piazza for most inquiries. There are tabs for different types of questions and discussions. The TAs will check into the Piazza page once a day on weekdays and respond to your questions within 24 hours. We encourage you to help your peers, but please do not post complete answers. We learn best when we have to figure things out for ourselves, with appropriate help when we get stuck.

If you have questions that you do not wish to share on Piazza, you may email the instructors or the TAs, using the TA email address (see above). For administrative inquiries please contact the TAs—you are much more likely to get a rapid response from them than from the instructors. Make sure that you use your Purdue email address and allow 24 hours for a response. Please include "AAE 251" and your section number in the subject line of all your emails.

If you have something that you would like to discuss that does not work well over email and may include components you do not want to share during student help sessions, please email the instructors directly to set up a one-on-one meeting or phone call. We are happy to try and answer questions about internships, careers, course selection, and even life in general.

Class and help sessions times, and assignment due dates are all in the Eastern Time Zone. Plan your work accordingly. Don't leave submissions to the last minute or second—plan for potential issues like unreliable internet connections.

To be considerate to the health, safety, and well-being of your fellow students and the teaching team, do not come to class if you are feeling ill, have a cold, or are running a fever. If you test positive for COVID-19, follow the guidelines found at the Protect Purdue website.

2 Course Description and Learning Outcomes and Objectives

Our purpose in this course is to deepen your understanding and excitement for aerospace engineering. You will work in teams over the semester to develop either a **space system** or an **aircraft** design. You will also have the opportunity to learn from your classmates about past, present, and future aerospace concepts through the "Vehicle of the Week" (VOTW) presentation. We will begin the semester by learning about the **process** of designing a system, from figuring out how to turn a vague idea into a detailed concept design, to how to communicate your solution clearly both in writing and orally. As you progress through your design project, we will introduce the **key aerospace engineering concepts, principles, and technical knowledge**.

A student who successfully fulfills the course requirements will have demonstrated the ability to:

Learning Outcome 1: Name and describe the parts of aerospace systems and their missions.

- LO 1.1 List and describe the terms used in the aerospace community to identify the geometry of an airfoil and wing.
- LO 1.2 List and describe the terms used in the aerospace community to identify the parts of an aircraft and their functions.
- LO 1.3 List and describe the terms used in the aerospace community to identify the parts of a launch vehicle and their functions.
- LO 1.4 List and describe the elements of an aircraft mission for different types of aircraft.

Learning Outcome 2: Create mission, needs, and requirements statements for complex aerospace systems.

- LO 2.1 Identify stakeholders and describe their roles and perspectives.
- LO 2.2 Write a mission statement for a system design project or program.

- LO 2.3 Describe the importance of proper requirements engineering to system development.
- LO 2.4 Establish the steps in stakeholder needs elicitation and generate needs statements.
- LO 2.5 Write proper needs statements for an engineering system.
- LO 2.6 Write proper requirements statements for an engineering system.
- LO 2.7 Identify and describe the hazards faced by an aerospace vehicle, their effects on the vehicle, and ways to mitigate these hazards.

Learning Outcome 3: Derive and apply equations that describe fluid flow to predict aerodynamic performance.

- LO 3.1 Identify the layers in the atmosphere.
- LO 3.2 Derive equations showing how temperature and pressure vary with altitude.
- LO 3.3 Identify and list the parameters that describe fluid flow (pressure, temperature, density, and velocity).
- LO 3.4 Derive and apply the relationships between the parameters that describe fluid flow (pressure, temperature, density, and velocity) to simple airfoil analysis and design problems.
- LO 3.5 Derive and apply Bernoulli's principle, the continuity equation, and the momentum equation to determine the lift acting on an aerodynamic surface.
- LO 3.6 Identify the parts of a wind tunnel.
- LO 3.7 List the two types of fluid flow and explain how they are replicated in a wind tunnel.
- LO 3.8 Define and explain the significance of the Reynolds number.
- LO 3.9 Use the Reynolds number to appropriately scale models for testing in wind tunnels.
- LO 3.10 Derive and apply the equations that describe laminar flow in a wind tunnel.
- LO 3.11 Identify the forces acting on an aerodynamic surface (lift, drag, thrust, gravity).
- LO 3.12 List the parameters that affect lift and drag on an aerodynamic surface, and describe their effects (angle of attack, density, velocity, and surface area).
- LO 3.13 Identify and describe the mechanisms through which the different types of subsonic drag arise (pressure, induced, and skin friction).
- LO 3.14 Explain the difference between finite and infinite airfoils. Explain why the infinite airfoil concept is useful.
- LO 3.15 Construct the theoretical drag polar plot.
- LO 3.16 Select and apply the appropriate equation to calculate the lift and drag on a finite airfoil given the lift-curve slope and drag polar for an infinite airfoil.
- LO 3.17 Use the drag polar plot to predict aircraft aerodynamic performance.
- LO 3.18 Use the drag polar to derive an expression for total drag as a function of velocity.
- LO 3.19 Read airfoil charts for infinite wings to find coefficients of lift, drag, and moment given the angle of attack and vice-versa

Learning Outcome 4: Derive and apply equations that describe the motions of bodies in space to design launch and orbital trajectories.

- LO 4.1 Categorize the different types of trajectories in space.
- LO 4.2 Derive the orbit (trajectory) equations.
- LO 4.3 Apply Kepler's laws and explain their significance.
- LO 4.4 Describe different types of orbital maneuvers and their application.
- LO 4.5 Derive and apply the delta-V requirement for a given maneuver.
- LO 4.6 Apply the vis-viva equation to analyze and design orbital maneuvers.
- LO 4.7 Describe the trade-offs involved in orbit selection and launch location.
- LO 4.8 Solve a Hohmann Transfer problem to analyze and design orbital maneuvers.
- LO 4.9 Calculate orbital parameters using launch geometry and geometry of conics.

Learning Outcome 5: Derive and apply equations that describe rocket thrust to predict rocket performance.

- LO 5.1 Derive the thrust equation for a rocket engine by applying the conservation of momentum.
- LO 5.2 Define the specific impulse of an engine.
- LO 5.3 Derive the ideal rocket equation by applying the conservation of momentum.
- LO 5.4 Apply the ideal rocket equation to calculate rocket design parameters.
- LO 5.5 Identify and describe over- and under-expansion and their effects on rocket performance.
- LO 5.6 List and describe the parts of a rocket engine and their functions.
- LO 5.7 Derive equations for the mass of propellant and initial mass of a rocket using the ideal rocket equation and the inert mass fraction.

Learning Outcome 6: Use system-level requirements and derive and apply appropriate equations to develop conceptual rocket designs

- LO 6.1 Determine and motivate the appropriate propellant(s) for a given rocket application.
- LO 6.2 Define and categorize launch vehicles based on their destination, payload, nature of propellants, and number of stages.
- LO 6.3 Identify and discuss the factors that reduce performance from that determined by the ideal rocket equation.
- LO 6.4 Discuss the key parameters in the design of a rocket (e.g., payload, delta-V, engines, propellant, feasibility, etc.).
- LO 6.5 Identify and demonstrate the need for multi-stage rockets.
- LO 6.6 Detail the methods and calculations used to design multi-stage rockets.
- LO 6.7 Determine the number of stages and amounts of propellants required to provide a given delta-V for a rocket application.
- LO 6.8 Determine the optimum split in delta-V for two-stage rockets.
- LO 6.9 Write MATLAB code to solve rocket staging problems.

Learning Outcome 7: Use system-level requirements and derive and apply appropriate equations to develop conceptual aircraft designs

- LO 7.1 Apply the aircraft sizing algorithm and appropriately estimate the weight fractions used in the algorithm to estimate aircraft takeoff mass.
- LO 7.2 Identify the various parameters used in determining aircraft takeoff mass.
- LO 7.3 Derive the equations of motion for an aircraft.
- LO 7.4 Derive the thrust required for steady, level, unaccelerated aircraft flight as a function of velocity.
- LO 7.5 Derive the thrust required for steady, level, unaccelerated aircraft flight as a function of weight and aerodynamic efficiency.
- LO 7.6 Derive the thrust available and maximum velocity for an aircraft.
- LO 7.7 Derive the power required for steady, level, unaccelerated aircraft flight.
- LO 7.8 Derive the power available and maximum velocity for an aircraft.
- LO 7.9 Derive the minimum thrust and minimum power required for steady, level, unaccelerated flight for an aircraft.
- LO 7.10 Use the maximum thrust (power) available to derive the maximum possible velocity at a given altitude for an aircraft.
- LO 7.11 Derive and apply the Breguet Range equation to jet and propeller aircraft.
- LO 7.12 Derive and apply the endurance equation to jet and propeller aircraft.
- LO 7.13 Describe parameters and flight strategies to increase aircraft range.
- LO 7.14 Describe parameters and flight strategies to increase aircraft endurance.
- LO 7.15 Analyze altitude effects on required thrust and power for air-breathing engines.
- LO 7.16 Categorize propulsion systems into air breathing and rocket engines.
- LO 7.17 Determine and motivate the appropriate engine for a given aircraft application.
- LO 7.18 Identify and list the stages in air-breathing engines and their functions.

3 How to Succeed in this Course

This course provides you multiple opportunities to prove mastery of the material. To succeed in this course, watch the lecture videos posted on Brightspace. Do so regularly—set up a calendar with planned study time for all your courses. Make sure to ask questions and participate in the on-campus and/or online discussions. You will work on weekly individual and team assignments. Don't wait until the last day to begin the assignments. Don't hesitate to ask for help. You can reach out to instructors and TA in class, via email or use the Piazza discussion board. Make use of all the opportunities provided to you in class to build your final grade. **Remember, submitting an incomplete homework is always better than not submitting anything at all.**

For the team assignments, contribute actively to your team. Your team should set up a regular discussion at least once a week. Do your share of the work in a timely manner and help others complete their tasks. Your teammates might not always agree with you. Be open

to new ideas and respect others' point of view. If you have problem with your teammates, for example, continuous absence of a member in group meetings, notify the instructors immediately. We are experienced in helping mediate these kinds of situations and happy to help.

4 Learning Resources, Technology, & Texts

4.1 Required Textbook

Anderson, **Introduction to Flight**, 8th Edition. McGraw Hill, 2015 or Anderson, **Introduction to Flight**, 9th Edition. McGraw Hill, 2022. *The 8th and 9th editions have some material that we will be using that is not in the older editions. However, you can probably get by with an older edition—if you are unsure, please ask.*

4.2 Recommended Textbooks

You do NOT need to buy these books!

Nicolai and Carichner, **Fundamentals of aircraft and airship design**, AIAA, 2010. *Nicolai is available online at Purdue Libraries at no cost.*

Vallado and McClain, **Fundamentals of astrodynamics and applications**, Springer, 2007.

4.3 Course Material and Assignments

The course is available on Brightspace. For each lecture, we will upload the video lecture, lecture notes, and (What We Learned Today) WWLT worksheet to Brightspace by the regular in-class lecture times. You can watch the videos at your own pace. If you have trouble accessing a video, let us know immediately. We will also be using Brightspace for team submissions.

We will be using Gradescope for individual work submissions. We will handle any regrade requests for these submissions through Gradescope only. See Brightspace for some guidelines on how to scan and upload your work to Gradescope.

You will be notified of any changes via email or a Brightspace announcement.

4.4 A.I. Tool Use Policy

In this course, the use of A.I. is allowed for purposes such as finding or creating additional or new exercises related to the course material. A.I. tools *cannot* be used for partially or entirely solving or answering homework or exam (ISA) questions.

Note: For the VOTW, Book Report, and Team Project report, we will use an A.I. checking tool to evaluate the extent of AI assistance used in creating the report. While you are welcome to use tools like ChatGPT or Grammarly to get tips on improving your writing, it is important to

rewrite the content in your own words after using such assistance. Copy-pasting directly or with minimal edits from these tools or using their suggestions verbatim may flag the report for excessive AI-generated content.

5 Assignments

Refer to the updated individual assignment files posted on Brightspace for detailed instructions and guidelines.

- **Team Project**: You will work in teams on the semester-long design project. You will either design an air or a space system, progressing via a series of structured milestone assignments and peer reviews. At the end of the semester, your team must submit a project report and a presentation video. We suggest you set up a plan to communicate with your teammates as soon as possible. Plan your meetings and submission of work. Your project grade is based on your report, presentation video, and buddy ratings. Note that we take buddy ratings seriously—meaning you must make sure to pull your weight on your team assignments.

- **Homework**: Homework will typically be due weekly. Pay close attention to due dates. Please submit your homework to Gradescope as a PDF file. Unless indicated otherwise, you may submit homework up to 24 hours late with 10% deducted from the final score. After that, your homework will not be accepted. We will drop the homework for which you got the lowest score—if you miss a homework, it will count as your lowest score.

  Regrade Policy: We will handle regrades through Gradescope. You will have one week from receiving your graded work to submit your requests. Before submitting a regrade request, go over the solutions document posted on Brightspace. Make sure to explain clearly where your work was overlooked, or how your different method is correct. You may submit one regrade request; if you believe the regrade is incorrect, please email the instructors for a final decision. Keep in mind that if you received any "generous" points, they may be removed in the regrade. Do not make excuses and be respectful in your regrade requests.

- **Individual Self Assessments**: Upon completion of each of the course modules, we will ask you to compile a self-assessment of your learning. This assessment will include a comprehensive summary of all the material we covered. We will post instructions on Brightspace closer to the assignment dates.

- **Book Report**: Over the course of the semester, we would like you to read at least one book related to aerospace engineering. A book report will be due at the end of the semester, and we will hold “book club” during the last week of classes. Refer to Brightspace for detailed instructions.

- **Vehicle of the Week (VOTW)**: In this assignment, you will learn about a wide range of aerospace systems, both by researching one with your team and preparing a presentation

and report about it, and by watching other teams' presentations. Refer to Brightspace for detailed instructions.

6 Evaluation

We will use the following weighting to calculate your final scores for this course.

Homework:	**15%**
Individual Self Assessments:	**30%**
Participation:	**5%**
Book Report:	**5%**
Team Project:	**35%**
VOTW:	**10%**
Bonus Percents:	**Up to 4% (added to final grade)**

Your final grade will be based on your final score and will be calculated according to the following scale. Instructors reserve the right to modify the grading scale.

A+	98.00 – 100.00%
A	94.00 – 97.99%
A-	90.00 – 93.99%
B+	87.00 – 89.99%
B	84.00 – 86.99%
B-	80.00 – 83.99%
C+	77.00 – 79.99%
C	74.00 – 76.99%
C-	70.00 – 73.99%
D+	67.00 – 69.99%
D	64.00 – 66.99%
D-	60.00 – 63.99%
F	<60.00%

Participation: You can earn this point by coming to class and responding to the daily attendance quiz question using iClicker Cloud. You do not need to attend every single class session to earn the maximum participation points. We assume and expect that you will miss some sessions—we are counting on you to skip class judiciously and not capriciously.

Bonus Percents: A total of 20 'Bonus Building Points' will account for the 4% added to the final grade. You can earn Bonus Building Points in two ways:

1. **Attend a Talk:** You can earn 2 Bonus Building Points per occasion by attending on-campus or online talks relevant to Aerospace Engineering. To receive your points for

attending a talk, you must upload to Brightspace five bullet points explaining what you learned, as well as a photo of yourself at the event, or a screenshot of the event showing your attendance. No photo, no points; no bullet points, no points—NO NEGOTIATION. If you are not sure whether a particular talk will be relevant, email the TA to check.

Note: The talks need to be related to an engineering topic. While yes, many of these companies are engineering companies, the seminars are recruitment events, so they will be covering very little engineering information. Some examples of talks which would count towards the assignment are if a visiting professor comes to give a talk on their research, or if an engineering organization on campus, such as AIAA, hosts a guest speaker to talk about engineering work that they do. If a company comes to campus to speak about a specific engineering topic (for instance, developing a new aircraft to reduce fuel emissions), then you could use this, but **if the talk is primarily to recruit, network, or give general information about the company, you cannot use it for the assignment.**

2. **Course Evaluation:** During the last two weeks of the course, you will be provided with an opportunity to evaluate this course and your instructor. Purdue uses an online course evaluation system. You will receive an official email from evaluation administrators with a link to the online evaluation site. You can earn 1 Bonus Building Point for submitting a proof of completed course evaluation. You must upload a screenshot showing you completed the survey for AAE251. Please do not share your responses to the survey.

You must make all your submissions for Bonus Building Points by the Friday of the Quiet Period (11:59 pm Eastern Time).

A grade of **incomplete** (I) will be given only in unusual circumstances. To receive an "I" grade, submit a written request to the instructor. The request must describe the circumstances, along with a proposed timeline for completing the coursework. Submitting a request does not ensure that an incomplete grade will be granted. If granted, you will be required to fill out and sign an "Incomplete Contract" form that will be turned in with the course grades. Any requests made after the course is completed will not be considered for an incomplete grade. Note that because we post solutions to all homework assignments, you cannot submit any homework for credit after each homework's deadline.

7 Course Schedule

The course schedule is available on Brightspace. You will be notified in case of changes.

In the event of a major campus emergency, course requirements, deadlines and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances. Here are ways to get information about changes in this course:

- Email the instructor or the TA
- Monitor the course site on Brightspace and your Purdue email

Notes taken during lectures are, however, generally considered to be "derivative works" of the instructor's presentations and materials, and they are thus subject to the instructor's copyright in such presentations and materials. **No individual is permitted to sell or otherwise barter notes, either to other students or to any commercial concern, for a course without the express written permission of the course instructor.** To obtain permission to sell or barter notes, the individual wishing to sell or barter the notes must be registered in the course or must be an approved visitor to the class. Course instructors may choose to grant or not grant such permission at their own discretion, and may require a review of the notes prior to their being sold or bartered. If they do grant such permission, they may revoke it at any time, if they so choose.

Purdue's student guide for academic integrity also has some useful information.